Database Management Systems: Assignment-11

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Q 1. PL/SQL Program for Prime Number.

Code:

```
DECLARE
    num NUMBER:=&num;
    i NUMBER;
    flag BOOLEAN:=TRUE;
BEGIN
    IF num<=1 THEN
        flag:=FALSE;
    ELSE
        FOR i IN 2 .. FLOOR(SQRT(num)) LOOP
            IF num MOD i = 0 THEN
                flag:=FALSE;
                EXIT;
            END IF;
        END LOOP;
    END IF;

    IF flag THEN
        DBMS_OUTPUT.PUT_LINE(num || ' is a prime number.');
    ELSE
        DBMS_OUTPUT.PUT_LINE(num || ' is NOT a prime number.');
    END IF;
END;
/
```

Output:

```
SQL> DECLARE
        num NUMBER:=17;
        i NUMBER;
        flag BOOLEAN:=TRUE;...
Show more...


17 is a prime number.


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.003
```

Q 2. PL/SQL Program for Reverse of a Number.

Code:

```
DECLARE
    num NUMBER:=&num;
    original NUMBER:=num;
    rev NUMBER:=0;
    digit NUMBER;
BEGIN
    WHILE num>0 LOOP
        digit:=MOD(num, 10);
        rev:=rev*10+digit;
        num:=TRUNC(num/10);
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Original Number: ' || original);
    DBMS_OUTPUT.PUT_LINE('Reversed Number: ' || rev);
END;
/
```

Output:

```
SQL> DECLARE
        num NUMBER:=12345;
        original NUMBER:=num;
        rev NUMBER:=0;...
Show more...


Original Number: 12345
Reversed Number: 54321


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008
```

Q 3. PL/SQL Program for Fibonacci Series.

Code:

```
DECLARE
    n NUMBER:=&n;
    a NUMBER:=0;
    b NUMBER:=1;
    c NUMBER;
    i NUMBER:=1;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Fibonacci Series:');

    WHILE i<=n LOOP
        DBMS_OUTPUT.PUT_LINE(a);

        c:=a+b;
        a:=b;
        b:=c;

        i:=i+1;
    END LOOP;
END;
/
```

Output:

```
Fibonacci Series:
0
1
1
2
3
5
8
13
21
34


PL/SQL procedure successfully completed.
```

Q 4. PL/SQL Program to Check Number is Odd or Even.

Code:

```
DECLARE
    num NUMBER:=&num;
BEGIN
    IF MOD(num, 2)=0 THEN
        DBMS_OUTPUT.PUT_LINE(num || ' is Even.');
    ELSE
        DBMS_OUTPUT.PUT_LINE(num || ' is Odd.');
    END IF;
END;
/
```

Output:

```
SQL> DECLARE
         num NUMBER:=27;
     BEGIN
         IF MOD(num, 2)=0 THEN...
Show more...


27 is Odd.


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006
```

Q 5. PL/SQL Program to Reverse a String.

Code:

```
DECLARE
    str VARCHAR2(100):='&str';
    rev VARCHAR2(100):='';
    i NUMBER;
BEGIN
    FOR i IN REVERSE 1 .. LENGTH(str) LOOP
        rev:=rev || SUBSTR(str, i, 1);
    END LOOP;

    DBMS_OUTPUT.PUT_LINE('Original String: ' || str);
    DBMS_OUTPUT.PUT_LINE('Reversed String: ' || rev);
END;
/
```

Output:

```
SQL> DECLARE
        str VARCHAR2(100):='HELLO WORLD';
        rev VARCHAR2(100):='';
        i NUMBER;...
Show more...


Original String: HELLO WORLD
Reversed String: DLROW OLLEH


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008
```

Q 6. Pl/SQL Program for Palindrome Number.

Code:

```
DECLARE
    num NUMBER:=&num;
    original NUMBER:=num;
    rev NUMBER:=0;
    digit NUMBER;
BEGIN
    WHILE num>0 LOOP
        digit:=MOD(num, 10);
        rev:=rev*10+digit;
        num:=TRUNC(num/10);
    END LOOP;

    IF rev=original THEN
        DBMS_OUTPUT.PUT_LINE(original || ' is a Palindrome Number.');
    ELSE
        DBMS_OUTPUT.PUT_LINE(original || ' is NOT a Palindrome Number');
    END IF;
END;
/
```

Output:

```
SQL> DECLARE
        num NUMBER:=121;
        original NUMBER:=num;
        rev NUMBER:=0;...
Show more...


121 is a Palindrome Number.


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.006
```

Q7. PL/SQL Program to Swap two Numbers.

Code:

```
DECLARE
    a NUMBER:=&a;
    b NUMBER:=&b;
    temp NUMBER;
BEGIN
    DBMS_OUTPUT.PUT_LINE('Before Swap: A=' || a || ', B=' || b);

    temp:=a;
    a:=b;
    b:=a;

    DBMS_OUTPUT.PUT_LINE('After Swap: A=' || a || ', B=' || b);
```

```
END;
/
```

Output:

```
SQL> DECLARE
         a NUMBER:=10;
         b NUMBER:=20;
         temp NUMBER;...
Show more...


Before Swap: A=10, B=20
After Swap: A=20, B=20


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.008
```

Q8. PL/SQL Program for Armstrong Number.

Code:

```
DECLARE
    num NUMBER:=&num;
    original NUMBER:=num;
    digit NUMBER;
    sum NUMBER:=0;
    n NUMBER;
BEGIN
    n:=LENGTH(num);

    WHILE num>0 LOOP
        digit:=MOD(num, 10);
        sum:=sum+POWER(digit, n);
        num:=TRUNC(num/10);
    END LOOP;

    IF sum=original THEN
        DBMS_OUTPUT.PUT_LINE(original || ' is an Armstrong Number.');
    ELSE
        DBMS_OUTPUT.PUT_LINE(original || ' is NOT an Armstrong Number.');
    END IF;
END;
/
```

Output:

```
SQL> DECLARE
        num NUMBER:=153;
        original NUMBER:=num;
        digit NUMBER;...
Show more...


153 is an Armstrong Number.


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.004
```

Q9. PL/SQL Program to Find Greatest of Three Numbers.

Code:

```
DECLARE
    a NUMBER:=&a;
    b NUMBER:=&b;
    c NUMBER:=&c;
    greatest NUMBER;
BEGIN
    IF a>=b AND a>=c THEN
        greatest:=a;
    ELSIF b>=a AND b>=c THEN
```

```
        greatest:=b;
    ELSE
        greatest:=c;
    END IF;

    DBMS_OUTPUT.PUT_LINE('Greatest Number= ' || greatest);
END;
/
```

Output:

```
SQL> DECLARE
        a NUMBER:=10;
        b NUMBER:=25;
        c NUMBER:=15;...
Show more...


Greatest Number= 25


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.011
```

Q10. PL/SQL Program to Print Patterns.

```
*

**

***

****

*****
```

Code:

```
BEGIN
    FOR i IN 1..5 LOOP
        DBMS_OUTPUT.PUT_LINE(RPAD('*', i, '*'));
    END LOOP;
END;
/
```

Output:

```
*
**
***
****
*****


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.005
```

```
* * * * *
  * * * *
    * * *
      * *
        *
```

Code:

```
BEGIN
    FOR i IN REVERSE 1..5 LOOP
        DBMS_OUTPUT.PUT_LINE(LPAD(' ', 5-i) || RPAD('*', i, '*'));
    END LOOP;
END;
/
```

Output:

```
*****
 ****
  ***
   **
    *


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.007
```

```
    *
   ***
  *****
 *******
*********
```

Code:

```
BEGIN
    FOR i IN 1..5 LOOP
        DBMS_OUTPUT.PUT_LINE(LPAD(' ', 5-i) || RPAD('*', 2*i-1, '*'));
    END LOOP;
END;
/
```

Output:

```
    *
   ***
  *****
 *******
*********


PL/SQL procedure successfully completed.

Elapsed: 00:00:00.007
```